

1. Title Page

Lab Name: Lab 13: Wireless LANs (WLAN) - WiFi and 802.11
Analysis Date: YYYY-MM-DD **Name:** [Your Name] **Lab Partners:**
[Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling
Labs (CML) - Personal / Free Edition

2. Background

Wireless Local Area Networks (WLANs) operate under the **IEEE 802.11 (WiFi)** standard. Key concepts include: 1. **IEEE 802.11 Frame Format:** Unlike Ethernet frames which use a 2-address format (Source and Destination), 802.11 frames use up to a **4-address format**. This is because frames are transmitted over a shared wireless medium to an intermediate **Access Point (AP)**, which acts as a bridge to the wired distribution system. 2. **Addressing Mapping:** The roles of the address fields (Address 1, Address 2, Address 3, Address 4) are determined by the To DS (To Distribution System) and From DS flags in the Frame Control field: - To DS = 0, From DS = 1 (Downlink from AP to Client): Address 1 is Destination (Client), Address 2 is BSSID (AP), Address 3 is Source (Sender). - To DS = 1, From DS = 0 (Uplink from Client to AP): Address 1 is BSSID (AP), Address 2 is Source (Client), Address 3 is Destination (Receiver). 3. **802.11 Frame Types:** Frames are classified into: - **Management Frames:** Beacons, probe requests/responses, association requests/responses. (Beacons are broadcast periodically by the AP to advertise the network SSID, rates, and capabilities). - **Control Frames:** RTS/CTS (Request/Clear to Send) to avoid collisions and solve the hidden terminal problem, and ACKs to confirm frame delivery. - **Data Frames:** Encapsulate the actual network layer payloads. 4. **WPA2 4-Way Handshake:** Wireless links are encrypted to prevent eavesdropping. When a client associates, it performs a **4-way cryptographic handshake** using EAPOL (EAP over LAN) key exchanges to verify the Pre-Shared Key (PSK) and generate temporal encryption keys (PTK and GTK).

In this lab, you will deploy a wireless network in CML using a WiFi Access Point and two WiFi Client nodes, perform packet captures on the wireless interface, and analyze management, control, data, and security handshake sequences.

3. Objective / Purpose

To configure a basic WLAN infrastructure in CML, capture and analyze 802.11 management frames (beacons), inspect 802.11 frame address fields and control flag maps, and analyze the WPA2 4-way cryptographic key exchange in Wireshark.

4. Topology Diagram

```
graph TD
    H0["Host H0 (WiFi Client)<br>IP: 192.168.1.10/24<br>SSID: CS457-WiFi"] --> |"802.11 Link"| WAP0["WiFi AP WAP0<br>SSID: CS457-WiFi<br>BSSID: Radio MAC"]
    H1["Host H1 (WiFi Client)<br>IP: 192.168.1.11/24<br>SSID: CS457-WiFi"] --> |"802.11 Link"| WAP0

    classDef client fill:#d4f1f9,stroke:#007b8b,stroke-width:2px;
    classDef ap fill:#fff2cc,stroke:#d6b656,stroke-width:2px;

    class H0,H1 client;
    class WAP0 ap;
```

Details: - 1x WiFi Access Point (<initials>-WAP0) broadcasting SSID CS457-WiFi. - 2x WiFi Clients (<initials>-H0 and <initials>-H1) associated with the AP.

5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 2x WiFi Client Nodes, 1x WiFi Access Point Node

6. Configuration / Methodology

WLAN Configuration Table

Device	CML Node Name	Interface	IP Address	SSID	Security Mode
Access Point	<initials>-WAP0	Radio0	192.168.1.1/24	CS457-WiFi	WPA2-PSK (AES)
Client 0	<initials>-H0	wlan0	192.168.1.10/24	CS457-WiFi	WPA2-PSK (AES)
Client 1	<initials>-H1	wlan0	192.168.1.11/24	CS457-WiFi	WPA2-PSK (AES)

Step-by-Step Configuration Guide

1. Access Point Setup (<initials>-WAP0)

Open the **Edit Config** or configuration pane of the WiFi AP node in CML: 1. Set the hostname to <initials>-WAP0 (e.g., KH-WAP0). 2. Set the wireless SSID to CS457-WiFi. 3. Set the encryption type to WPA2-Personal (PSK) with AES. 4. Configure the Pre-Shared Key (password) to wirelesspass123. 5. Assign a static IP address to the AP's local interface: 192.168.1.1/24.

2. Client Host Setup (<initials>-H0 and <initials>-H1)

On each client host node, open the **Edit Config** tab to connect to the wireless network: 1. Configure Host <initials>-H0 hostname and wireless association parameters: ``bash # Configure wlan0 interface
ip addr add 192.168.1.10/24 dev wlan0

```
# Associate with the CML WiFi AP
iwconfig wlan0 essid CS457-WiFi key s:wirelesspass123
2. Configure Host <initials>-H1` hostname and wireless association parameters:
bash ip addr add 192.168.1.11/24 dev wlan0
iwconfig wlan0 essid CS457-WiFi key s:wirelesspass123 ``
```

Step-by-Step Traffic Generation & Execution:

1. **Start Packet Capture:** Right-click the wireless radio link or AP node interface, select **Packet Capture**, and click **Start**.
2. **Verify Client Association:** Open Host H0's console and verify connection status:

```
iwconfig wlan0
```

(Confirm that the Access Point MAC address is displayed, indicating successful association).

3. **Generate Traffic:** Ping Client H1 from Client H0:

```
ping -c 4 192.168.1.11
```

4. **Trigger 4-Way Security Handshake:** To capture the initial key negotiation, disconnect and reconnect Host H1 while capture is running:

```
# On Client H1: Disconnect and reconnect
ifconfig wlan0 down
sleep 2
ifconfig wlan0 up
iwconfig wlan0 essid CS457-WiFi key s:wirelesspass123
```

5. **Download Capture:** Stop the CML packet capture and download the .pcap capture file to your local computer.
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7. Wireshark Analysis and Questions

Open your downloaded packet capture in Wireshark and answer the following questions:

Part A: 802.11 Management Frames (Beacons)

1. Locate a **Beacon frame** sent by the AP (<initials>-WAP0).
 - What is the Source MAC address (the transmitter)?
 - What is the BSSID of the wireless network? Does this match the AP's MAC?
2. Inspect the **Beacon parameters** in the packet details pane:
 - What SSID is advertised?
 - What WPA/WPA2 security requirements (e.g. Robust Secure Network - RSN group/pairwise cipher suites) are listed in the parameters?

Part B: 802.11 Frame Addressing & Control Flags

3. Locate an **802.11 Data frame** containing your ping traffic (ICMP request) sent from Client H0 to Client H1.
 - Expand the **Frame Control** field in the IEEE 802.11 header. What are the binary/hex values of the To DS and From DS flags?
 - Based on these flag states, why are these values set (what is the directional path of the packet)?
4. Inspect the address fields in the header of this data frame:
 - What is the MAC address value for:
 1. Address 1 (Receiver Address / RA)
 2. Address 2 (Transmitter Address / TA)
 3. Address 3 (Source/Destination Address depending on DS flags)
 - Map these addresses to the actual CML nodes in your topology (H0, H1, or WAP0).
5. Explain why an IEEE 802.11 wireless frame header requires up to **four** MAC address fields, whereas a standard Ethernet II wired frame header only requires **two**. (Discuss the bridging role of the AP).

Part C: WPA2 Security and the 4-Way Handshake

6. Locate the EAPOL key negotiation packets during Host H1's reconnection.
 - How many EAPOL packets are exchanged to complete the **4-Way Handshake**?
 - What type of keys (PTK, GTK, PMK) are dynamically generated during this handshake?
 - Why is the 4-Way Handshake necessary (what does it protect against in a wireless environment)?
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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything]

Discovery Method: [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

9. Conclusion & Analysis

Summarize your findings. Contrast the operations of 802.11 wireless networks with standard 802.3 wired Ethernet, detailing the role of management frames in network advertising, 3/4-address mapping over bridged access points, and the necessity of EAPOL handshakes for link-layer security.

10. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- A single PDF report containing:
 - Your written answers to the 6 Wireshark analysis questions in Section 7.
 - A screenshot of your active CML topology showing the running hosts (<initials>-H0, <initials>-H1) and AP (<initials>-WAP0).
 - A screenshot of Host H0's console showing the output of `iwconfig wlan0`, confirming association.
 - An annotated screenshot of your Wireshark capture window showing the WPA2 4-Way Handshake EAPOL key exchange.